PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Life under Covid-19 for LGBT+ people in the UK: Systematic review of UK research on the impact of COVID-19 on sexual and gender minority populations
AUTHORS	McGowan, Victoria; Lowther, Hayley; Meads, Catherine

VERSION 1 – REVIEW

REVIEWER	Perone, Angela K. University of Michigan
REVIEW RETURNED	18-Apr-2021

GENERAL COMMENTS	This research is so important, particularly given the dearth of research and data collection generally on LGBT+ people and COVID-19. I have several minor suggestions to strengthen the paper as described below: 1. On page 5: rephrase the sentence about gay men having promiscuous sexual behavior to avoid stigmatization. Instead focus on sexual behavior without invoking “promiscuous” which has many negative historical connotations and associations, particularly in the LGBT+ community. One possible solution would be to use the wording from the study cited (e.g. Some gay men have reported casual sex during the pandemic, which may increase the risk of COVID-19). 2. Be consistent with wording for LGBT and LGBT+. There are some places referring to reports for LGB populations, and this makes sense to use for that reason. However, there are places throughout the paper that use both LGBT and LGBT+ that seem to be interchangeable. 3. I suggest eliminating the comparison between institutional racism with institutional homophobia. There are major differences between these concepts, and without diving into the nuances and/or discussing intersectionality more intentionally throughout the paper, the paper risks undercutting the larger argument, which is that SOGI data is very much needed. Invoking institutional homophobia is still helpful – but more effective without analogizing to race.

REVIEWER	Nunn, Amy S Brown University
REVIEW RETURNED	19-Apr-2021

GENERAL COMMENTS	This is an interesting article that provides a review of COVID among SOGI populations in the UK. I recommend publishing it. I have few suggested line edits, but 2 big picture comments.

	The author's tone about NGO reports is condescending. Those reports are the backbone of your article and seem to provide really important information that the authors present with a lot of detail, even as the scientific community has fallen short. I suggest changing the tone. The second thing is, the conclusion ends with question. This is anathema to a conclusion. I suggest changing from questions to statements.

REVIEWER	Beckham, S. Wilson Johns Hopkins University
REVIEW RETURNED	21-Apr-2021

GENERAL COMMENTS	Throughout: Please consider using the term “equity” and “health inequities” instead of “health inequalities.” Use “trans men”, “trans women”, “transgender men”, “transgender women” and “trans people” rather than “transmen” and “transwomen. “The latter forms are used purposely in transphobic discourse to categorize trans women outside of the category of women, and adopted by others without malice, as I assume the authors here used it. Introduction: The literature cited in the introduction focuses primarily on biological differences—endocrine conditions, reproductive steroids, natural testosterone levels among sexual minority women, exogenous hormone use among trans people, HIV co-infection, and behavioral differences (“promiscuous sexual behavior” in gay men). While there may be something to biological differences between LGBT people and cisgender, heterosexual people, it’s clear that the primary drivers of health inequities for LGBT people are stigma and discrimination. See the minority stress model, first theorized by Virginia Brooks about sexual minority women, then adapted and expanded by Ilan Mayer. This posits that is it societal causes—stigma and discrimination, including internalized stigma, that leads to poorer health outcomes, less access to health care, etc. Remove discriminatory policies, stigma from healthcare providers and others, etc., and LGBT health inequities disappear. I encourage the authors to rework the introduction to include these social and structural factors in addition to the hypothesized biological factors. Furthermore, some of the language used is stigmatizing and victim-blaming. Please review and revise. • page 1, ln 47: ”Mental health is already poorer in the LGBT communities” without the context of this being caused by stigma and discrimination from society makes it sound as though being LGBT is inherently dangerous to mental health. Anti-LGBT groups pounce on such things to further stigmatize. Try “LGBT communities already face mental health disparities due to stigma and discrimination…” • page 6, ln 2: “promiscuous sexual behavior.” This is judgmental. Having multiple sexual partners is not inherently dangerous to one’s health. Reword to something like “People who have multiple

	sexual partners without taking necessary precautions may be at greater risk of acquiring COVID-19." Ln 55-56, "lesbians may have…" should be "Sexual minority women have…" since the systematic review covered research among lesbians and bisexual women hormone levels, not just lesbians. That said, I wouldn't include this at all given the cautious interpretation from the sys review, and instead focus on the social and structural determinants of health inequities in LGBT people, as discussed above. I certainly agree that routine data collection around SOGI is warranted! Methods I appreciate the statement about how authors are LGBT or allies, and it was reviewed by an LGBT org. Did any trans people review it? Results Table 1: Please define abbreviations (e.g. PTNS, BAME); also indicate what the * is for, and check thoroughly for mistakes (there's a random = column 2) Table 1 & Table 2: The listing of numbers of men/boys, women/girls, and trans is odd, and further perpetuates the idea that (binary) trans people are Other and Not Women and Not Men. Please consider listing these categories (as far as the data is available in the included reports, and when it is not, indicate so): Cis women Cis men Trans women Trans men Nonbinary/genderqueer/Another gender I understand that often, trans people are not disaggregated into men, women, and nonbinary people. In that case, please indicate "trans (not specified)" or something similar. In most studies that include trans people, the vast majority are trans women. It is useful to disaggregate whenever possible so we can learn more about the differential needs of trans men and nonbinary people. This is especially true if there is any substance to the hypothesis that sex (whether hormones or something else) matters in COVID outcomes. For example, if testosterone is a key factor in the differences, then trans men taking testosterone might be more at risk that cis women, and trans women on androgen suppressants and estrogen might be less at risk than cis men. Page 10, line 1 "journal article manuscript"; do you mean unpublished manuscript intended for publication in a peer-reviewed journal? The quote about funding on the 1st page of results makes a good point, but this isn't the space for it. This could be used as a point (without the full quote) in the Discussion about funding (and lack thereof).

	Page 10: final paragraph: “The two studies by academic groups30,31 did have ethical approval, but it is not clear whether the other nine projects sought this.” Remove the clause about the projects’ ethical approval. I’m not familiar with how the UK generally applies ethical review, but as they were not academic studies and not research per se, they would not need ethical approval. Table 2: why is Barnardo’s in this table, as they don’t have a comparator? And why is Lancashire in Table 3, for the same reason? The Results writeup as a whole could use more fleshing out with summary or synthesis. How do the reports speak to each other and tell us an overall (if scant) picture of COVID’s impact on LGBT people in the UK? The section on reports without comparators certainly deserves more space. The absolute effect of the COVID-19 pandemic on a marginalized community is valuable information on its own and does not necessarily need a comparator. To devalue these reports to the point of summarizing them in one very brief paragraph is unfortunate. Synthesize the results more—what can be said about physical health overall, mental health overall, etc., based on what is available, even if it is all grey lit. It’s all there is, after all. Discussion The discussion covers key points but is too informal. I suggest rewording the unanswered questions to statements as recommendations for policymakers and researchers. For example, “X phenomenon is worth further consideration.” Emphasize the point about lack of SOGI data in routine data collection—that’s a key takeaway here. If it were in routine data collection for COVID stats for the UK, the disparities could be analyzed.

VERSION 1 – AUTHOR RESPONSE

Peer reviewer 1 – Dr. Angela K. Perone, University of Michigan

	This research is so important, particularly given the dearth of research and data collection generally on LGBT+ people and COVID-19. I have several minor suggestions to strengthen the paper as described below:	We thank the peer reviewer for their comments on this.
1	On page 5: rephrase the sentence about gay men having promiscuous sexual behavior to avoid stigmatization. Instead focus on sexual behavior without invoking “promiscuous” which has many negative historical connotations and associations, particularly in the LGBT+ community. One possible solution would be to use the wording from the study cited (e.g. Some gay men have reported casual sex during the pandemic, which may increase the risk of COVID-19).	Thank you for this comment, we have amended the paper as you suggest.
2	Be consistent with wording for LGBT and LGBT+. There are some places referring to reports for LGB populations, and this makes sense to use for that reason. However,	Many thanks for pointing this out. What we have tried to do is use the specific initialisation (LGB/LGBT+/LGBTQ+ etc)

	there are places throughout the paper that use both LGBT and LGBT+ that seem to be interchangeable.	where specified by the source material, and where not specified, we have now used the generic LGBT+ initialisation.
3	I suggest eliminating the comparison between institutional racism with institutional homophobia. There are major differences between these concepts, and without diving into the nuances and/or discussing intersectionality more intentionally throughout the paper, the paper risks undercutting the larger argument, which is that SOGI data is very much needed. Invoking institutional homophobia is still helpful – but more effective without analogizing to race.	Thank you, that's very helpful. We have removed the institutional racism part.

Peer reviewer 2 - Dr. Amy S Nunn, Brown University

1	This is an interesting article that provides a review of COVID among SOGI populations in the UK. I recommend publishing it.	We thank the peer reviewer for their comments on this.
2	I have few suggested line edits*, but 2 big picture comments. The author's tone about NGO reports is condescending. Those reports are the backbone of your article and seem to provide really important information that the authors present with a lot of detail, even as the scientific community has fallen short. I suggest changing the tone.	We thank the peer reviewer for this comment, which has made us think about this issue. On the one hand, these included studies were never intended to be academic documents, so to systematically review them as if they are academic documents can make us sound condescending. On the other hand, without these we would have had nothing to systematically review. Hopefully the thorough treatment of the included studies, for example in the tables, demonstrates that we take these reports seriously. We have attempted to change the language to make it more realistic and factual and less condescending.
3	The second thing is, the conclusion ends with question. This is anathema to a conclusion. I suggest changing from questions to statements.	Thank you, we have changed all questions into statements in the discussion section.

* We have been unable to see any specific line edits when we log on to the resubmission page. Hopefully the in-depth peer review from Peer Reviewer 3 will have address any that Peer Reviewer 2 may have made.

Peer reviewer 3 - Dr. S. Wilson Beckham, Johns Hopkins University

	Throughout: Please consider using the term “equity” and “health inequities” instead of “health inequalities.”	We agree that you make a good point so in the text we have tried to use both terms as accurately as possible. (Using as a guide: Mariana C. Arcaya, Alyssa L. Arcaya & S. V. Subramanian (2015) Inequalities in health: definitions, concepts, and theories, Global Health Action, 8:1, 27106, DOI: 10.3402/ gha.v8.27106)
	Use “trans men”, “trans women”, “transgender men”, “transgender women” and “trans people” rather than “transmen” and “transwomen. “The latter forms are used purposely in transphobic discourse to categorize trans women outside of the category of women, and adopted by others without malice, as I assume the authors here used it.	Thank you for highlighting this and we have amended the text as advised.
	Introduction: The literature cited in the introduction focuses primarily on biological differences—endocrine conditions, reproductive steroids, natural testosterone levels among sexual minority women, exogenous hormone use among trans people, HIV co-infection, and behavioral differences (“promiscuous sexual behavior” in gay men). While there may be something to biological differences between LGBT people and cisgender, heterosexual people, it’s clear that the primary drivers of health inequities for LGBT people are stigma and discrimination. See the minority stress model, first theorized by Virginia Brooks about sexual minority women, then adapted and expanded by Ilan Mayer. This posits that is it societal causes—stigma and discrimination, including internalized stigma, that leads to poorer health outcomes, less access to health care, etc. Remove discriminatory policies, stigma from healthcare providers and others, etc., and LGBT health inequities disappear. I encourage the authors to rework the introduction to include these social and structural factors in addition to the hypothesized biological factors.	We have now changed the ‘promiscuous sentence – see peer reviewer 1 responses. We agree that social and structural factors are important regarding health inequities. We note the peer reviewers comments around removing discriminatory policies (we highlight these in yellow), but we know of no proof that this happens in practice. We have added a short paragraph in the background section on the impact of minority stress on physical and mental health.
	Furthermore, some of the language used is stigmatizing and victim-blaming. Please review and revise. • page 1, ln 47: ”Mental health is already poorer in the LGBT communities” without the context of this being caused by stigma and discrimination from society makes it sound as though being LGBT is inherently dangerous to mental health. Anti-LGBT groups pounce on such things to further stigmatize. Try “LGBT communities already face	Association is not causation. We know that mental health is poorer in the LGBT communities, it does not necessarily follow that it is being caused by being LGBT. We have rephrased the sentence.

	mental health disparities due to stigma and discrimination…"	
	• page 6, ln 2: "promiscuous sexual behavior." This is judgmental. Having multiple sexual partners is not inherently dangerous to one's health. Reword to something like "People who have multiple sexual partners without taking necessary precautions may be at greater risk of acquiring COVID-19."	We have now changed the 'promiscuous sentence – see peer reviewer 1 responses.
	Ln 55-56, "lesbians may have…" should be "Sexual minority women have…" since the systematic review covered research among lesbians and bisexual women hormone levels, not just lesbians. That said, I wouldn't include this at all given the cautious interpretation from the sys review, and instead focus on the social and structural determinants of health inequities in LGBT people, as discussed above.	We have amended the sentence to sexual minority women. We have added material on social and structural determinants, as outlined above.
	I certainly agree that routine data collection around SOGI is warranted!	We thank the peer reviewer for this comment.
	Methods I appreciate the statement about how authors are LGBT or allies, and it was reviewed by an LGBT org. Did any trans people review it?	Hart Gables did peer review the paper; the gender identity of the people reading were not disclosed to us.
	Results: Table 1: Please define abbreviations (e.g. PTNS, BAME); also indicate what the * is for, and check thoroughly for mistakes (there's a random = column 2)	We thank the peer reviewer for the detail of this, all now amended.
	Table 1 & Table 2: The listing of numbers of men/boys, women/girls, and trans is odd, and further perpetuates the idea that (binary) trans people are Other and Not Women and Not Men. Please consider listing these categories (as far as the data is available in the included reports, and when it is not, indicate so): Cis women Cis men Trans women Trans men Nonbinary/genderqueer/Another gender	We try very hard to reflect what is actually reported in the included study, and in Barnardo's this is how they reported it. However, we do agree with the peer reviewer so have amended Table 1 to *45% cis male, 45% cis female, 2% trans male plus trans female plus non-binary.* For table 2 we have added male and female to the headings to reflect that it is male and female people who can be trans or cis.
	I understand that often, trans people are not disaggregated into men, women, and nonbinary people. In that case, please indicate "trans (not specified)" or something similar. In most studies that include trans people, the vast majority are trans women. It is useful to disaggregate whenever possible so we can learn more about the differential needs of trans men and nonbinary people. This	We have added 'gender not specified' in Table 1. The Outlife survey does not specify male or female in the trans and cis comparator results. We do agree that the

	is especially true if there is any substance to the hypothesis that sex (whether hormones or something else) matters in COVID outcomes. For example, if testosterone is a key factor in the differences, then trans men taking testosterone might be more at risk that cis women, and trans women on androgen suppressants and estrogen might be less at risk than cis men.	difference in rates between male and female would be very useful for the reasons you suggest, and have added this lack of data to the discussion.
	Page 10, line 1 “journal article manuscript”; do you mean unpublished manuscript intended for publication in a peer-reviewed journal?	Yes we do, text amended.
	The quote about funding on the 1st page of results makes a good point, but this isn’t the space for it. This could be used as a point (without the full quote) in the Discussion about funding (and lack thereof).	We have now moved quote to discussion section.
	Page 10: final paragraph: “The two studies by academic groups30,31 did have ethical approval, but it is not clear whether the other nine projects sought this.” Remove the clause about the projects’ ethical approval. I’m not familiar with how the UK generally applies ethical review, but as they were not academic studies and not research per se, they would not need ethical approval.	Research is defined as the creation of new knowledge and/or the use of existing knowledge in a new and creative way so as to generate new concepts, methodologies and understandings. Service evaluations are designed to answer the question “What standard does this service achieve?” We think that the nine projects by the charities do fall within the remit of research, rather thn service delivery, because they are generating new knowledge. Ethical approval could be sought for these projects, because there is a possible ethical concern that a member of the public using the service provided by a charity may worry that continuing service delivery might be contingent on them completing the survey and on the responses they give. So we have kept in the clause about ethical approval.
	Table 2: why is Barnardo’s in this table, as they don’t have a comparator? And why is Lancashire in Table 3, for the same reason?	Barnardo’s has the cis comparator – the term is now in the heading. For the Lancashire survey, the historical comparator is implicit

		in how the questions are termed i.e 'less' or 'reduction'.
	The Results writeup as a whole could use more fleshing out with summary or synthesis. How do the reports speak to each other and tell us an overall (if scant) picture of COVID's impact on LGBT people in the UK?	The text of this section has now been enlarged to give a better picture overall of the study results.
	The section on reports without comparators certainly deserves more space. The absolute effect of the COVID-19 pandemic on a marginalized community is valuable information on its own and does not necessarily need a comparator. To devalue these reports to the point of summarizing them in one very brief paragraph is unfortunate. Synthesize the results more—what can be said about physical health overall, mental health overall, etc., based on what is available, even if it is all grey lit. It's all there is, after all.	The difficulty we have is that most of the evidence has no comparator, so we do not know if the results are to do with the Covid pandemic and lockdown, or would have happened anyway. We have enlarged this section as much as we can.
	Discussion: The discussion covers key points but is too informal. I suggest rewording the unanswered questions to statements as recommendations for policymakers and researchers. For example, "X phenomenon is worth further consideration." Emphasize the point about lack of SOGI data in routine data collection—that's a key takeaway here. If it were in routine data collection for COVID stats for the UK, the disparities could be analyzed.	Questions have now been changed to statements. We have already mentioned lack of routine data collection in discussion, conclusion and in abstract.

VERSION 2 – REVIEW

REVIEWER	Perone, Angela K. University of Michigan
REVIEW RETURNED	14-Jun-2021

GENERAL COMMENTS	This is a useful article that can guide future research.